

Overview

Data analysis is essential to evidence-based medicine, guiding both clinical decisions and research interpretation.¹ However, many clinicians face technical barriers due to limited training in statistics and data science.² This can lead to inefficiencies, dependence on external analysts, and slower translation of insights into practice. Recent advances in AI, especially large language models (LLMs), offer a promising solution, making data analysis more accessible by enabling natural language interactions with clinical data.

Large language models (LLMs) are transforming healthcare, supporting tasks from informatics research³ to AI-assisted clinical decision-making.⁴ These models are especially powerful in code generation for data analysis.⁵ AI code-writing agents built on LLMs can automate tasks like statistical testing and machine learning, streamlining clinical research. However, high-performing LLMs typically require cloud-based inference, raising data privacy concerns. Even de-identified data may carry re-identification risks⁶—posing ethical and regulatory challenges.

Methods

To address these challenges, we developed ChatDA (**Figure 1**), an AI agent for clinical tabular data analysis that does not rely on code generation. Inspired by recent successes in agentic tool use, ChatDA operates a specialized toolkit (TableMage, a custom low-code data science package) to ensure:

- Consistent and interpretable results with reduced error risk
- No direct data access by the LLM
- Tool-based analysis returns only population-level insights to the LLM

We evaluated the performance of ChatDA, OpenAI Advanced Data Analysis, Open Interpreter, and Data Interpreter⁵ on a range of non-machine learning (non-ML) and supervised machine learning (ML) tasks (**Figure 2**). Inspired by InfiAgent-DABench,⁷ an existing data analysis agent benchmark, we curated a novel benchmark comprising 100 questions across 10 datasets addressing InfiAgent-DABench's excessive technical specificity and lack of question continuity. We additionally performed an agent benchmarking study on 11 tabular ML tasks.

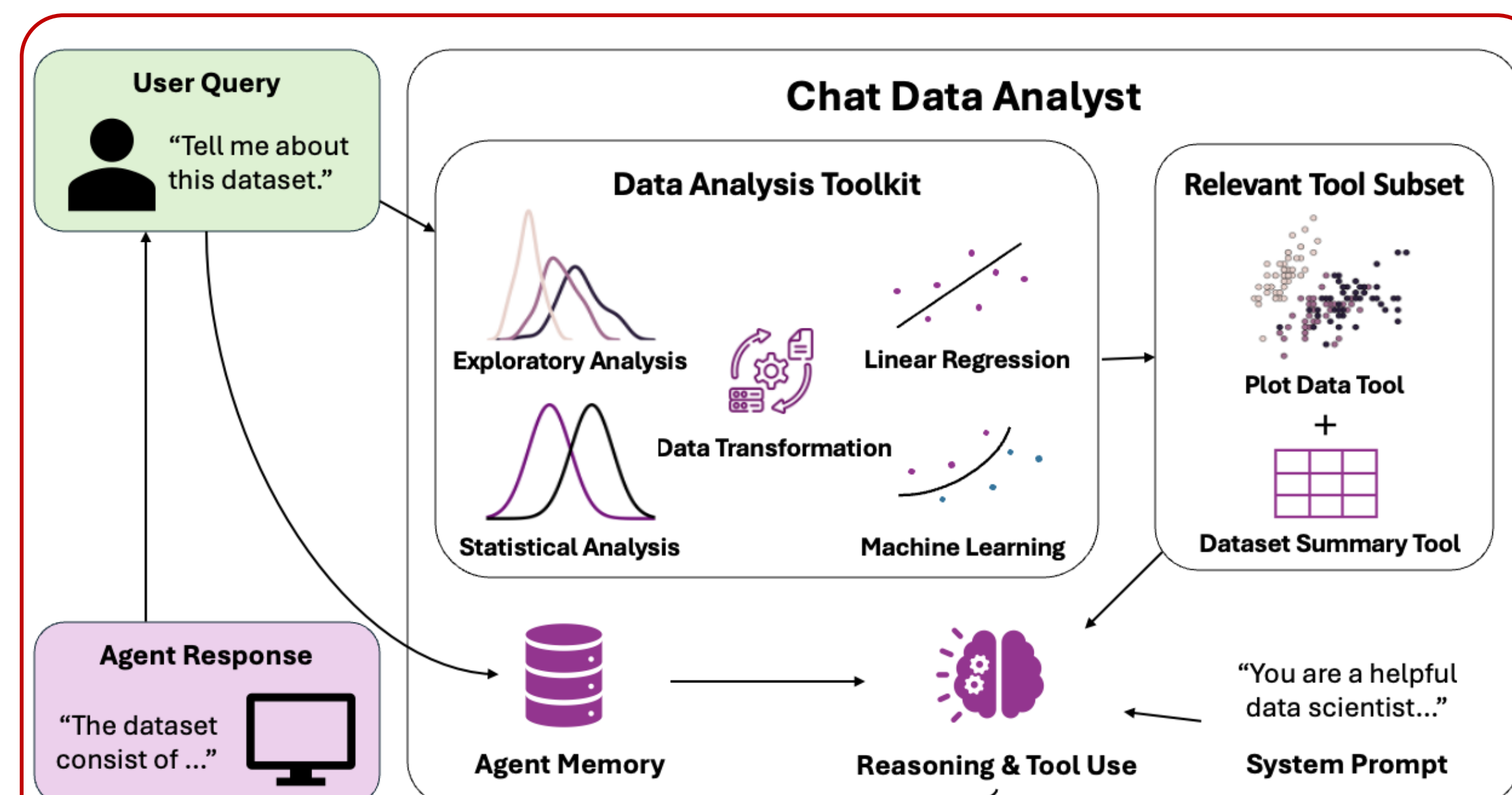


Figure 1. ChatDA design. ChatDA operates via a custom toolkit built on TableMage, a novel, low-code clinical data analysis software package.

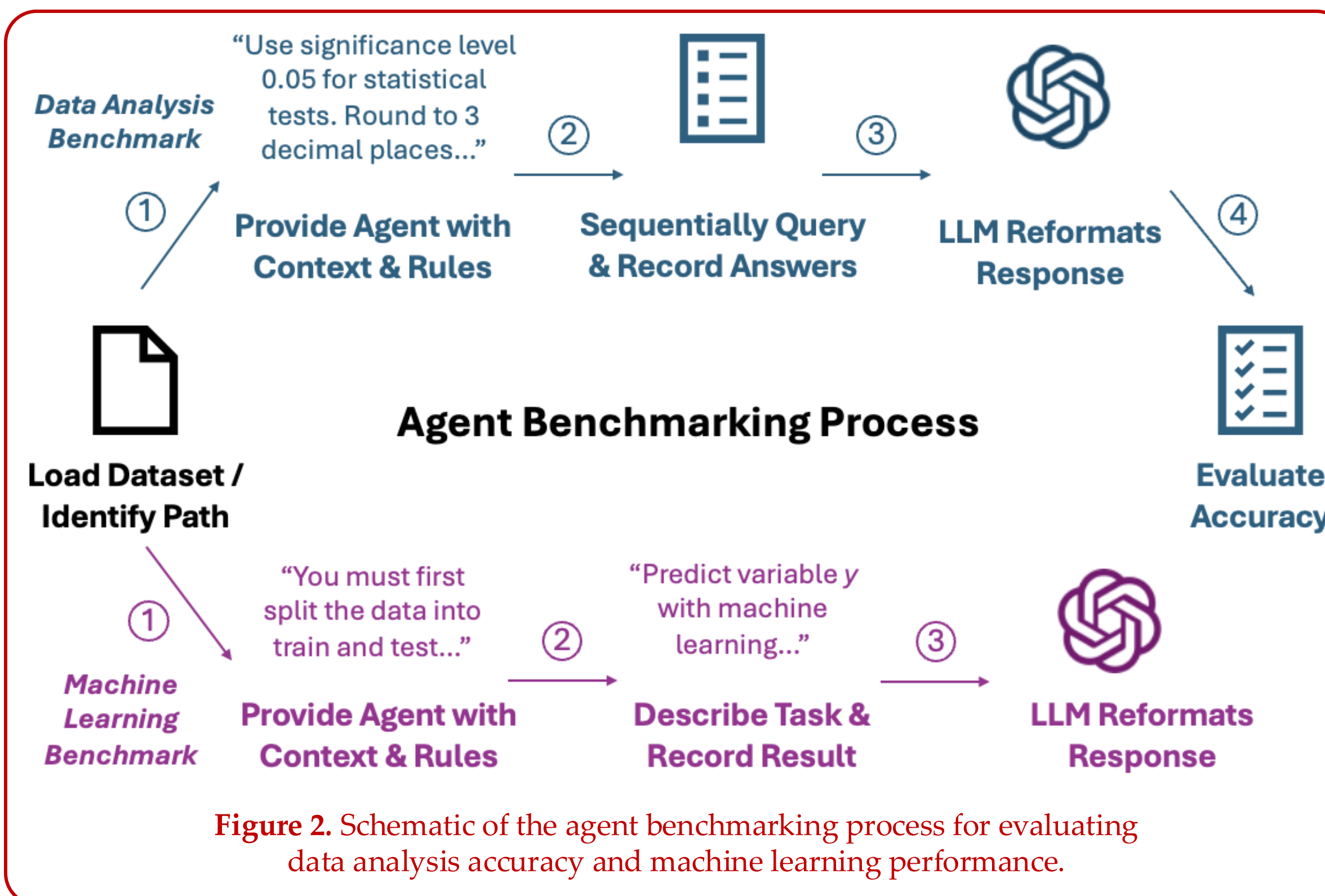


Figure 2. Schematic of the agent benchmarking process for evaluating data analysis accuracy and machine learning performance.

Results

Non-ML benchmark (Figure 3)

- ChatDA outperformed all agents with 10–18% higher accuracy
- ChatDA demonstrated superior output stability, evidenced by narrower confidence intervals across runs
- ChatDA outperformed other agents in retaining data transformations across analysis steps, likely due to its integrated toolkit that preserves feature changes automatically

ML benchmark

- ChatDA outperformed OpenAI Advanced Data Analysis in 10 out of 11 tasks (i.e., achieved a lower average test RMSE for regression or a higher average test AUC for classification)
- ChatDA outperformed Open Interpreter in 6 out of 11 tasks
- ChatDA outperformed Data Interpreter in 9 out of 11 tasks

Privacy considerations

- ChatDA preserves privacy with cloud LLMs by disabling raw LLM data access, sharing only population-level results through its tools

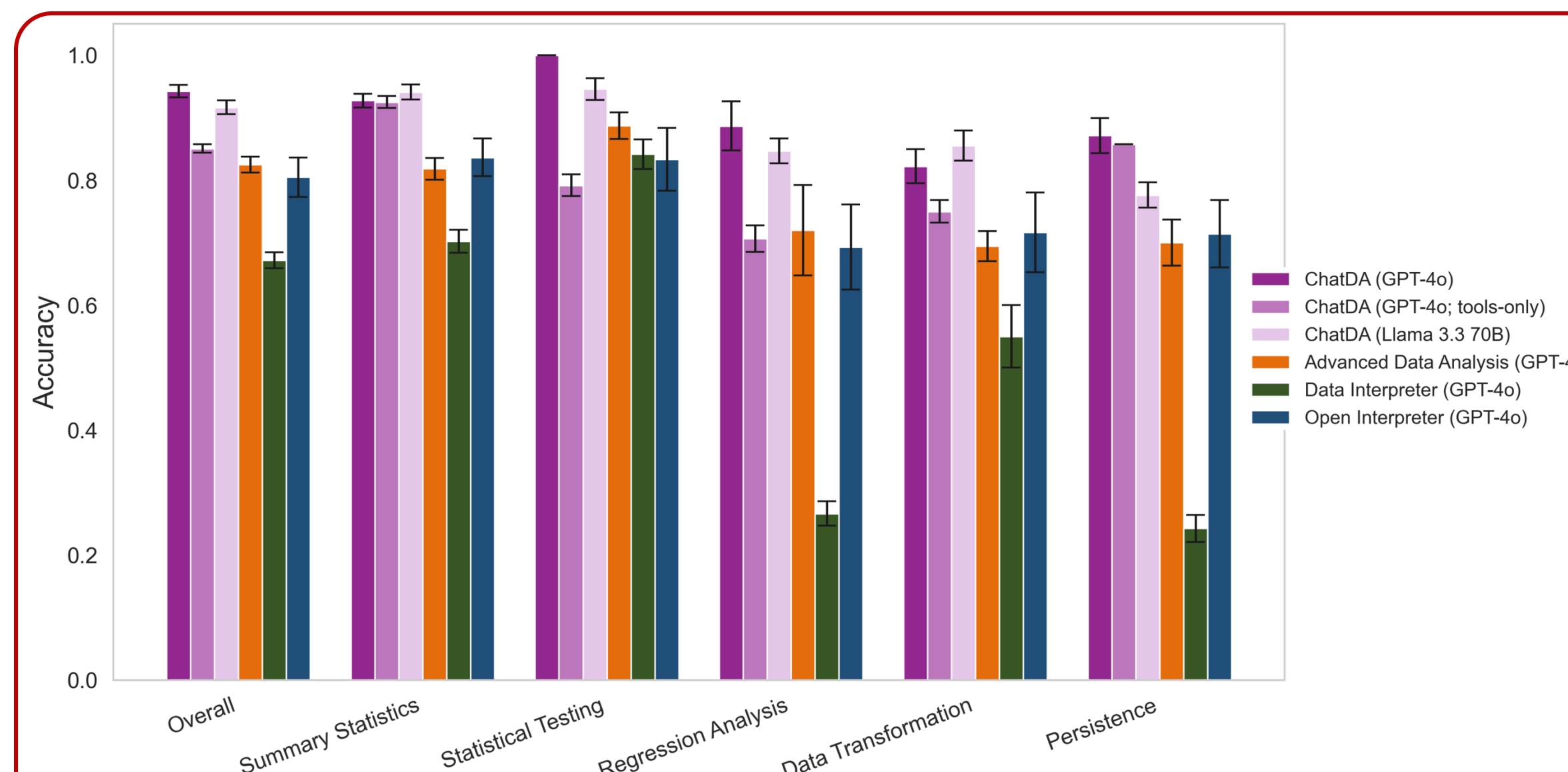


Figure 3. Accuracies of ChatDA, OpenAI Advanced Data Analysis, Data Interpreter, and Open Interpreter on the 100-question agentic data analysis (non-ML) benchmark

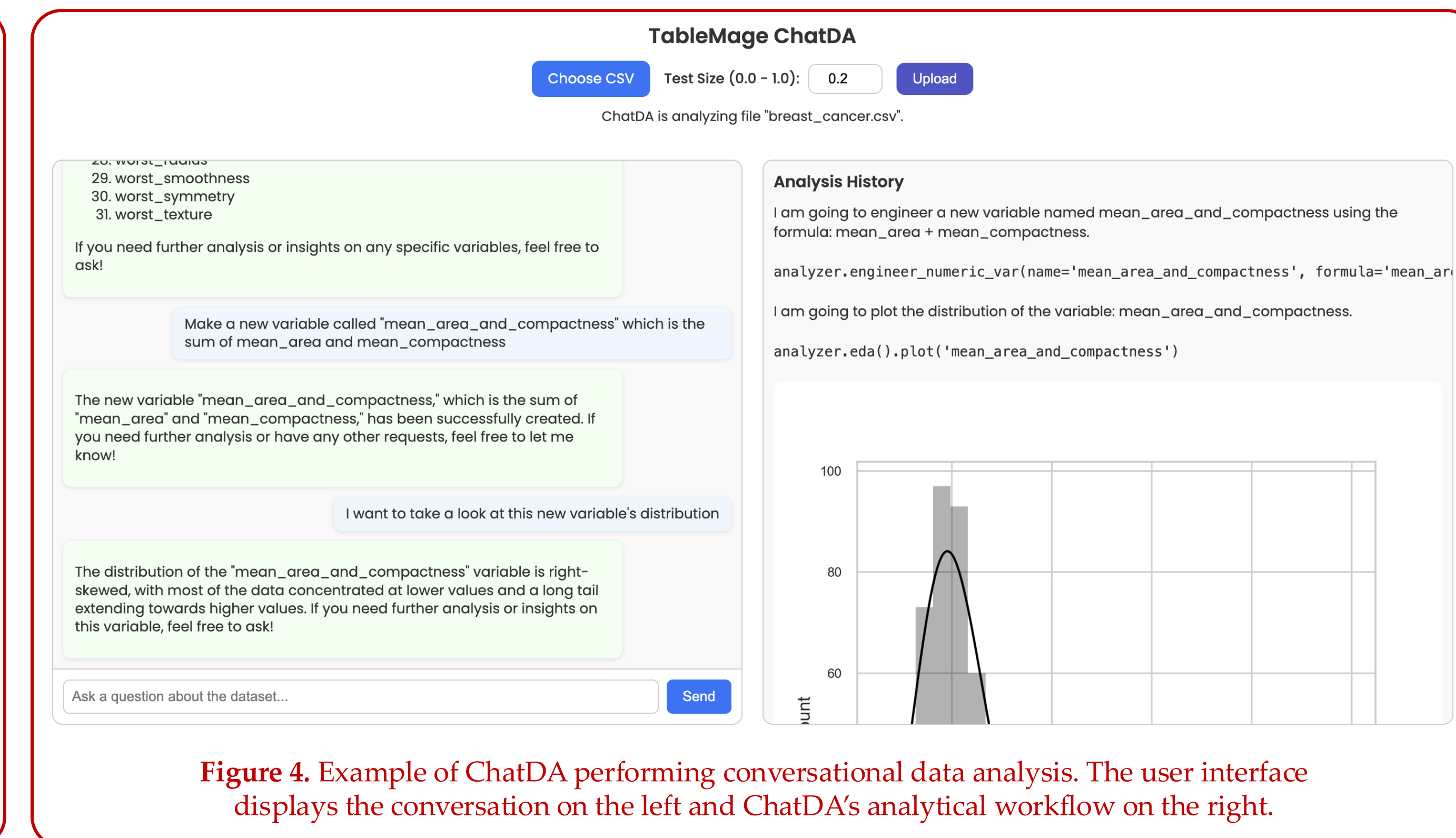


Figure 4. Example of ChatDA performing conversational data analysis. The user interface displays the conversation on the left and ChatDA's analytical workflow on the right.

Closing

We introduced ChatDA, a tool-calling agent designed for no-code, conversational clinical data analysis. We demonstrated that ChatDA achieves performance comparable to state-of-the-art code-writing data analysis agents, despite operating without unrestricted access to the underlying data. By restricting individual-level data access, ChatDA reduces the risk of compromising patient data privacy.

ChatDA empowers healthcare professionals to extract meaningful insights and advance evidence-based clinical practice through secure, AI-driven workflows that lower traditional barriers to data analysis. It has the potential to transform quantitative clinical research by accelerating discovery, broadening access to data-driven insights, and strengthening the foundation for clinical decision-making in an increasingly data-rich environment.

Data & Code Availability

ChatDA, along with a user interface (**Figure 4**), is available directly through the open source TableMage Python package (<https://github.com/ajy25/TableMage>). TableMage is available for download through the Python Package Index. All experiments in this study were performed with TableMage v0.1.0a1.

Datasets and scripts to reproduce similar results are organized in a publicly available repository (<https://github.com/ajy25/TableMage-Analysis>).

References

1. Reis, J. Evidence-Based Medicine: Era of Big Data. *Biomed. J. Sci. Tech. Res.* 41, 32901–32904 (2022).
2. Gigerenzer, G., Gaissmaier, W., Kurz-Milcke, E., Schwartz, L. M. & Woloshin, S. Helping Doctors and Patients Make Sense of Health Statistics. *Psychol. Sci. Public Interest* 8, 53–96 (2007).
3. Tayebi Arasteh, S. et al. Large language models streamline automated machine learning for clinical studies. *Nat. Commun.* 15, 1603 (2024).
4. Wang, D. & Zhang, S. Large language models in medical and healthcare fields: applications, advances, and challenges. *Artif. Intell. Rev.* 57, 299 (2024).
5. Hong, S. et al. Data Interpreter: An LLM Agent For Data Science. Preprint at <https://doi.org/10.48550/arXiv.2402.18679> (2024).
6. Sarkar, A. R., Chuang, Y.-S., Mohammed, N. & Jiang, X. De-identification is not enough: a comparison between de-identified and synthetic clinical notes. *Sci. Rep.* 14, 29669 (2024).
7. Hu, X. et al. InfiAgent-DABench: Evaluating Agents on Data Analysis Tasks. in *Proceedings of the 41st International Conference on Machine Learning 19544–19572 (PMLR, 2024)*.